

Problem 1: Forward Kinematics of a Differential Drive Robot

A differential drive mobile robot has a wheel radius of $r = 0.1 \text{ m}$ and a distance between the two wheels (track width) of $L = 0.5 \text{ m}$. At a given instant, the left wheel angular velocity is $\omega_l = 4 \text{ rad/s}$ and the right wheel angular velocity is $\omega_r = 6 \text{ rad/s}$.

The robot's current orientation (heading angle) is $\theta = \frac{\pi}{4} \text{ rad}$ (45°).

1. Calculate the linear velocity v of the robot's center of mass.
2. Calculate the angular velocity ω of the robot.
3. Calculate the instantaneous velocities in the global coordinate frame ($\dot{x}$ and $\dot{y}$).

Solution:

1. Linear Velocity (v):

The linear velocity of the center of mass is the average of the linear velocities of the right and left wheels ($v_r = r \cdot \omega_r$ and $v_l = r \cdot \omega_l$).

$$v = \frac{v_r + v_l}{2} = \frac{r(\omega_r + \omega_l)}{2}$$

$$v = \frac{0.1 \cdot (6 + 4)}{2} = \frac{1.0}{2} = \mathbf{0.5 \text{ m/s}}$$

2. Angular Velocity (ω):

The angular velocity of the robot chassis is determined by the speed difference between the wheels divided by the track width.

$$\omega = \frac{v_r - v_l}{L} = \frac{r(\omega_r - \omega_l)}{L}$$

$$\omega = \frac{0.1 \cdot (6 - 4)}{0.5} = \frac{0.2}{0.5} = \mathbf{0.4 \text{ rad/s}}$$

3. Global Velocities ($\dot{x}$, $\dot{y}$):

Using the kinematic rotation matrix mapping the robot frame to the global frame:

$$\dot{x} = v \cdot \cos(\theta) = 0.5 \cdot \cos\left(\frac{\pi}{4}\right) = 0.5 \cdot 0.7071 = \mathbf{0.354} \text{ m/s}$$

$$\dot{y} = v \cdot \sin(\theta) = 0.5 \cdot \sin\left(\frac{\pi}{4}\right) = 0.5 \cdot 0.7071 = \mathbf{0.354} \text{ m/s}$$

Problem 2: Inverse Kinematics (Trajectory Tracking)

A supervisor wants the same robot ($r = 0.1$ m, $L = 0.5$ m) to trace a circular path of radius $R = 2.0$ m at a constant linear speed of $v = 1.0$ m/s.

Calculate the required angular velocities (ω_r and ω_l) for the right and left wheels.

Solution:

First, we find the required robot angular velocity ω to maintain the circular path:

$$\omega = \frac{v}{R} = \frac{1.0}{2.0} = 0.5 \text{ rad/s}$$

Now, we set up the inverse kinematics equations for wheel velocities based on v and ω :

$$v_r = v + \frac{\omega \cdot L}{2} = 1.0 + \frac{0.5 \cdot 0.5}{2} = 1.0 + 0.125 = 1.125 \text{ m/s}$$

$$v_l = v - \frac{\omega \cdot L}{2} = 1.0 - \frac{0.5 \cdot 0.5}{2} = 1.0 - 0.125 = 0.875 \text{ m/s}$$

Convert linear wheel velocities to angular wheel velocities ($\omega = \frac{v}{r}$):

$$\omega_r = \frac{v_r}{r} = \frac{1.125}{0.1} = \mathbf{11.25} \text{ rad/s}$$

$$\omega_l = \frac{v_l}{r} = \frac{0.875}{0.1} = \mathbf{8.75} \text{ rad/s}$$

Part 2: Mobile Robot Dynamics

Dynamics considers the masses, moments of inertia, and forces/torques required to achieve accelerations.

Problem 3: Acceleration Force and Wheel Torque

A differential drive mobile robot has a total mass $m = 20 \text{ kg}$ and a moment of inertia $I = 0.8 \text{ kg}\cdot\text{m}^2$ about its center of mass. The wheel radius is $r = 0.1 \text{ m}$ and track width is $L = 0.5 \text{ m}$.

The robot needs to accelerate straight forward uniformly from a standstill to a velocity of 2 m/s over a time span of $t = 4 \text{ seconds}$. Assuming no friction or slip, calculate the motor torque (τ) required at *each* wheel during this acceleration phase.

Solution:

Step 1: Find the required linear acceleration (a).

$$a = \frac{v_f - v_i}{t} = \frac{2 - 0}{4} = 0.5 \text{ m/s}^2$$

Since the robot is traveling in a straight line, the angular acceleration (α) of the chassis is 0 rad/s^2 .

Step 2: Calculate total force (F_{total}) required for linear motion.

Using Newton's Second Law:

$$F_{total} = m \cdot a = 20 \text{ kg} \cdot 0.5 \text{ m/s}^2 = 10 \text{ N}$$

Step 3: Distribute the force to each wheel.

Because the motion is perfectly symmetric and straight, the total force is split equally between the two driving wheels:

$$F_{wheel} = \frac{F_{total}}{2} = \frac{10}{2} = 5 \text{ N per wheel}$$

Step 4: Calculate the required motor torque (τ).

Torque is force multiplied by the wheel radius:

$$\tau = F_{wheel} \cdot r = 5 \text{ N} \cdot 0.1 \text{ m} = \mathbf{0.5 \text{ N} \cdot \text{m}}$$

Each motor must continuously supply **0.5 N·m** of torque for 4 seconds.

Problem 4: Rotational Dynamics (Pure Spin)

The same robot ($m = 20 \text{ kg}$, $I = 0.8 \text{ kg} \cdot \text{m}^2$, $r = 0.1 \text{ m}$, $L = 0.5 \text{ m}$) needs to spin on the spot (pure rotation, $v=0$) with an angular acceleration of **$\alpha = 2 \text{ rad/s}^2$** .

Calculate the torques required from the right (τ_r) and left (τ_l) wheel motors.

Solution:

Step 1: Calculate the total torque ($T_{chassis}$) needed to rotate the chassis.

Using Newton's Second Law for rotation:

$$T_{chassis} = I \cdot \alpha = 0.8 \text{ kg} \cdot \text{m}^2 \cdot 2 \text{ rad/s}^2 = 1.6 \text{ N} \cdot \text{m}$$

Step 2: Relate chassis torque to wheel forces.

For a pure counterclockwise spin, the right wheel pushes forward with force $+F$ and the left wheel pulls backward with force $-F$. The couple creates a moment about the center:

$$T_{chassis} = \left(F \cdot \frac{L}{2} \right) - \left(-F \cdot \frac{L}{2} \right) = F \cdot L$$

$$1.6 = F \cdot 0.5 \implies F = \frac{1.6}{0.5} = 3.2 \text{ N}$$

So, $F_r = 3.2 \text{ N}$ (forward) and $F_l = -3.2 \text{ N}$ (backward).

Step 3: Convert wheel forces to motor torques.

$$\tau_r = F_r \cdot r = 3.2 \cdot 0.1 = \mathbf{0.32 \text{ N} \cdot \text{m}}$$

$$\tau_l = F_l \cdot r = -3.2 \cdot 0.1 = \mathbf{-0.32 \text{ N} \cdot \text{m}}$$

To achieve this spin, the right motor must apply **0.32 N·m** of torque, while the left motor must apply a braking/reverse torque of **-0.32 N·m**.